SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Pseudo code Writing Task (Algorithm Design)

Course Code:	DSA0308	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	2024	**Date:**	

Code of Conduct

I B V Chaitanya Reddy-192425376) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: chaitanya

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4
Machine Translation, Interlingua, Statistical Approaches	Translation Workflow Algorithm	Q5

Annexure A – Pseudocode Writing Task (Algorithm Design)

1. Reference Resolution Using Multiple Constraints

Design an algorithm to perform **Reference Resolution** by identifying pronouns and selecting the most appropriate antecedent using gender, number, recency, and semantic compatibility.

Apply your algorithm to the discourse:

"Meena gave Kavya a laptop because she needed one for her project. She thanked her and started using it."

Perform the following steps:

1. Identify all entities and referring expressions.
2. Generate possible antecedents for each pronoun.
3. Apply gender, number, recency, and semantic constraints.
4. Select the most appropriate antecedent.
5. Generate the resolved discourse.

Finally, explain how your algorithm avoids incorrect resolution when multiple antecedents have similar grammatical properties.

2. Algorithm for Entity Chains and Discourse Coherence

Design an algorithm to identify **entity chains** in a discourse and use them to evaluate text coherence.

Apply your algorithm to the following text:

"Anjali bought a new laptop. The laptop had a powerful processor. She installed Python on it. Later, Anjali used the computer to develop a machine learning application."

Perform the following steps:

1. Identify all entities and referring expressions.
2. Link expressions referring to the same entity.
3. Construct entity chains.
4. Calculate or estimate the continuity of entities across sentences.
5. Determine whether the discourse is coherent.

Finally, explain how breaking an entity chain could reduce discourse coherence.

3. Algorithm for Identifying Discourse Relations

Design an algorithm to identify **discourse relations** between consecutive sentences using discourse markers and contextual meaning.

Apply your algorithm to:

"The server crashed unexpectedly. As a result, users could not access the application. The technical team restarted the server. After that, the system became available again."

Your algorithm should identify relations such as:

- Cause–Effect
- Sequence
- Problem–Solution
- Elaboration

Perform the following steps:

1. Divide the discourse into individual discourse units.
2. Identify explicit discourse markers.
3. Analyze the semantic relationship between units.
4. Assign an appropriate discourse relation.
5. Construct a discourse structure representation.

Finally, justify how the identified relations make the discourse logically coherent.

4. Algorithm for Dialogue Act Classification

Design an algorithm to recognize **Dialogue Acts** in a conversation.

Apply your algorithm to:

User: "Hello, I need help with my assignment."
Agent: "Sure, what problem are you facing?"
User: "I do not understand machine learning."
Agent: "I can explain the basic concepts to you."
Classify each utterance into one of the following:

- Greeting
- Request
- Question
- Inform
- Offer
- Confirmation

Perform the following steps:

1. Preprocess each utterance.
2. Extract relevant linguistic features.
3. Identify the communicative intention.
4. Assign a dialogue act.
5. Generate the dialogue-act sequence.

Finally, explain how dialogue-act recognition helps a conversational agent select an appropriate next response.

5. Algorithm for Dialogue State Tracking
Design an algorithm for a **Conversational Agent** that maintains dialogue state while collecting information from a user.
Apply your algorithm to:
User: "I want to book a flight."
Agent: "Where would you like to travel?"
User: "I want to go to Delhi."
Agent: "From which city?"
User: "From Chennai."
Your algorithm should maintain the following slots:
Departure City
Destination City
Travel Date
Number of Passengers
Perform the following steps:

1. Initialize the dialogue state.
2. Identify information provided in each utterance.
3. Update the corresponding slot.
4. Identify missing information.
5. Generate the next appropriate question.

Finally, explain how dialogue state tracking improves coherence in a multi-turn conversation.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach

Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1	4	3	3	4	3	17	83%
2	4	4	3	3	3	17	
3	4	3	3	3	3	16	
4	4	3	3	3	4	17	
5	3	3	4	3	3	16	

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

1. Reference Resolution Using Multiple Constraints

Aim:

To identify pronouns in a discourse and resolve them to the most appropriate antecedent using gender, number, recency, and semantic compatibility constraints.

Algorithm:

1. Identify all entities and pronouns in the discourse.
2. Generate candidate antecedents for each pronoun.
3. Check gender compatibility.
4. Check number compatibility.
5. Apply recency and semantic constraints.
6. Select the best antecedent and replace the pronoun.

Pseudocode:

```
BEGIN
Extract entities and pronouns
FOR each pronoun
   Find candidate antecedents
   Check gender compatibility
   Check number compatibility
   Apply recency constraint
   Apply semantic constraint
   Select best antecedent
END FOR
Replace pronouns with antecedents
Display resolved discourse
END
```

Input Discourse:

"Meena gave Kavya a laptop because she needed one for her project. She thanked her and started using it."

Entities and Referring Expressions

- Meena

- Kavya
- Laptop
- she
- her
- it

Antecedent Selection:

Pronoun	Antecedent
she (needed one)	Kavya
her (project)	Kavya
She (thanked her)	Kavya
her	Meena
it	Laptop

Resolved Discourse:

"Meena gave Kavya a laptop because Kavya needed one for Kavya's project. Kavya thanked Meena and started using the laptop."

Explanation: The algorithm avoids incorrect resolution by checking gender, number, recency, and semantic meaning together instead of relying on only one constraint.

2. Algorithm for Entity Chains and Discourse Coherence

Aim:

To identify entity chains and evaluate discourse coherence.

Algorithm:

1. Identify entities and references.
2. Link references to the same entity.
3. Construct entity chains.
4. Measure continuity across sentences.
5. Evaluate coherence.

Pseudocode:

BEGIN

Identify entities

Find referring expressions

Link references to entities

Create entity chains

Measure continuity across sentences

Determine coherence score

Display result

END

Input Text:

"Anjali bought a new laptop. The laptop had a powerful processor. She installed Python on it. Later, Anjali used the computer to develop a machine learning application."

Entity Chains:

Chain 1: Anjali

- Anjali
- She
- Anjali

Chain 2: Laptop

- Laptop
- The laptop
- it
- computer

ContinuityAnalysis:

Entities continue across all sentences without interruption.

CoherenceResult:

The discourse is coherent because major entities are consistently referenced.

Explanation:

Breaking an entity chain makes it difficult to track the topic, reducing discourse coherence.

3. Algorithm for Identifying Discourse Relations

Aim:

To identify discourse relations between discourse units.

Algorithm:

1. Split discourse into units.
2. Identify discourse markers.
3. Analyze semantic relations.
4. Assign discourse relation.
5. Build discourse structure.

Pseudocode:

```
BEGIN
Split text into discourse units
Identify discourse markers
Analyze semantic relationships
Assign discourse relation
Construct discourse structure
Display result
END
```

Input Discourse:

"The server crashed unexpectedly. As a result, users could not access the application. The technical team restarted the server. After that, the system became available again."

Discourse Units

1. Server crashed.
2. Users could not access application.
3. Team restarted server.
4. System became available.

Relations Identified

Units	Relation
1 → 2	Cause–Effect
2 → 3	Problem–Solution
3 → 4	Sequence
4	Elaboration

Discourse Structure:

Server Crash

↓

Users Cannot Access

↓

Server Restart

↓

System Available

Explanation:

These relations create logical flow and improve text coherence.

4. Algorithm for Dialogue Act Classification

Aim:

To classify each utterance into an appropriate dialogue act.

Algorithm:

1. Preprocess utterances.
2. Extract linguistic features.
3. Identify communicative intention.
4. Assign dialogue act.
5. Generate dialogue sequence.

Pseudocode:

```
BEGIN
Read utterances
Preprocess text
Extract features
Determine intention
Assign dialogue act
Generate sequence
Display result
END
```

Conversation:

User: "Hello, I need help with my assignment."

Agent: "Sure, what problem are you facing?"

User: "I do not understand machine learning."

Agent: "I can explain the basic concepts to you."

Dialogue Act Classification

Utterance	Dialogue Act
Hello	Greeting
I need help with my assignment	Request
What problem are you facing?	Question
I do not understand machine learning	Inform
I can explain the basic concepts to you	Offer

Dialogue Act Sequence

Greeting → Request → Question → Inform → Offer

Explanation

Dialogue-act recognition helps conversational agents choose appropriate responses and maintain natural interaction.

5. Algorithm for Dialogue State Tracking

Aim:

To maintain dialogue state while collecting user information.

Algorithm:

1. Initialize slots.
2. Extract information from utterances.
3. Update slots.
4. Identify missing slots.
5. Ask the next relevant question.

Pseudocode:

```
BEGIN
Initialize slots as empty
Read user input
Extract information
Update slots
Check missing slots
Generate next question
Display state
END
```

Conversation:

User: "I want to book a flight."

Agent: "Where would you like to travel?"

User: "I want to go to Delhi."

Agent: "From which city?"

User: "From Chennai."

Dialogue State Table

Slot	Value
Departure City	Chennai
Destination City	Delhi
Travel Date	Empty
Number of Passengers	Empty

Missing Information

- Travel Date
- Number of Passengers

Next Appropriate Question

"What date would you like to travel?"

Explanation:

Dialogue State Tracking maintains conversation context, prevents repeated questions, and improves multi-turn dialogue coherence.

Result:

The algorithms successfully demonstrate:

- Reference Resolution
- Entity Chain Construction
- Discourse Relation Identification
- Dialogue Act Classification
- Dialogue State Tracking

These techniques improve Natural Language Processing systems by increasing understanding, coherence, and conversational effectiveness.